AnchorMTM: Spatially Anchored Merge Tree Maps for Time-Varying Scalar Fields

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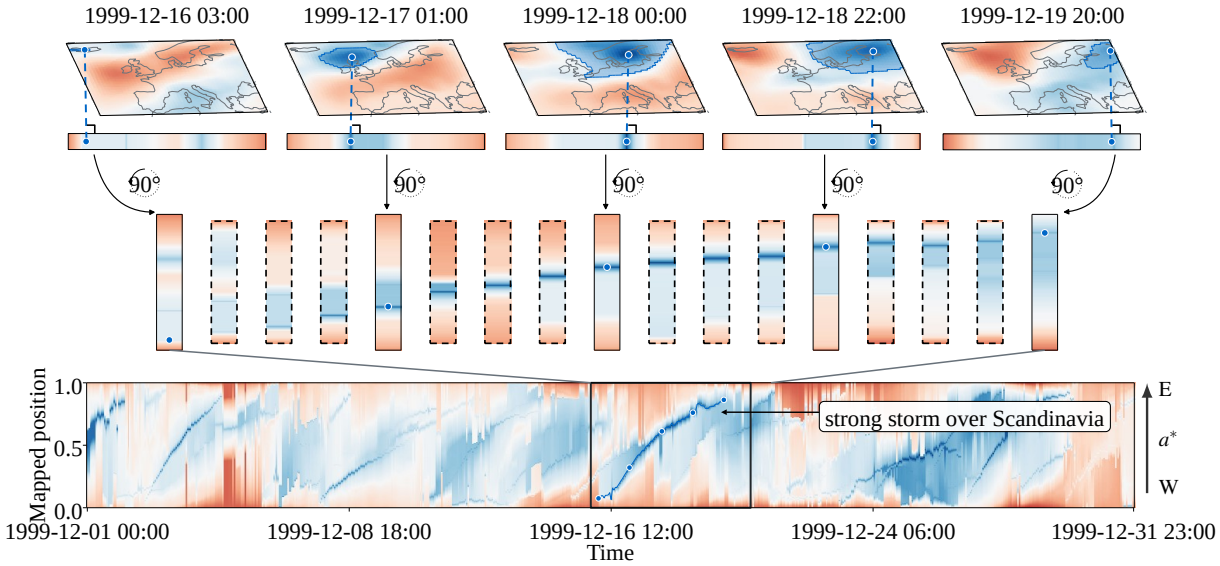


Figure 1: Storms temporal overview ($\varepsilon = 5\%$): 744 hourly ERA5 sea-level-pressure fields, December 1999. Each frame's merge tree becomes a one-dimensional profile, rotated and stacked along time. The vertical axis is one fixed spatial direction a^* (arrow, west to east), so the rising track is a storm moving east, from the open Atlantic to Scandinavia; the snapshots identify the matched extremum, and blue outlines show its leaf-arc support. The map keeps every nesting and merge level within the 5% budget.

Abstract

Merge tree maps summarize a time-varying scalar field in a single static image: each time step becomes a one-dimensional profile of its merge tree, and the profiles are stacked along time, so that nested features appear as nested bands. The vertical axis of these maps, however, encodes tree order or relative distance rather than location, so an analyst can see how features nest but not where they are or where they move. We present AnchorMTM, a spatially anchored merge tree map whose vertical axis is one fixed spatial direction. Anchoring exposes a structural conflict: an exact profile keeps every subtree contiguous, whereas the projected locations of sibling subtrees may interleave. We quantify the displacement this conflict forces through three layout-independent lower bounds, for exact topology, for features of finite width, and for a bounded change of merge levels, and we build the map around them. A leaf-order search admits orders through a closed-form merge-error test, a capped quadratic program places the extrema, and a single merge budget controls how much merge-level fidelity may be traded for position. On five planar and two volumetric sequences, the selected orders attain the width-aware bound on 1,335 of 1,624 planar frames; on the real-data sequences, 56–98% of the order error of existing merge tree maps is not forced by the tree; and at zero merge error, AnchorMTM lowers the mean worst displacement of TMTM and ST-MTM by 73–96%, while ST-MTM retains better neighborhood preservation. The bounds make the positional price of a spatial axis measurable before a map is drawn.

Index Terms: Time-varying scalar fields, merge tree maps, topology-based visualization, spatial anchoring, lower bounds, constrained layout.

1 Introduction

Time-varying scalar fields record how quantities such as sea-level pressure, aerosol concentration or vorticity evolve over a spatial domain. Analysts study them through features, such as low-pressure systems, pollution episodes or shed vortices, and ask three questions about each: how it relates to the others, where it is, and how it moves. Animation keeps every time step in its spatial context but makes distant times hard to compare [45, 51]. Static overviews therefore assign one display axis to time and compress space into the other, so that a whole sequence fits into one image [2]. What that second axis means decides which of the three questions the overview can answer. The Hovmöller diagram of atmospheric science keeps a physical coordinate such as longitude and shows an eastward-moving system as a slanted streak [25], but it averages the field over the remaining coordinates, so neighboring features blend and their hierarchy disappears.

Merge tree maps answer the first question. A merge tree records

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how sublevel sets connect as the threshold rises: its leaves are the extrema, and its internal nodes are the levels at which regions merge [13, 21]. Temporal merge tree maps (TMTM) draw the tree of each time step as a one-dimensional profile whose minima are the extrema and whose barriers are the merge levels, and they stack the profiles into one image [30]. Spatiotemporal merge tree maps (ST-MTM) additionally arrange the features by their pairwise distances and temporal coherence [10]. In both, the vertical axis follows the tree, temporal alignment or relative distance; neither fixes an origin or a direction in space. A storm that crosses the North Atlantic within a few days can therefore stay at the same height of the map, and the reader sees how it connects to neighboring lows but not where it is or which way it moves.

We give this axis a spatial meaning. Fixing one direction a^* in the domain assigns every extremum a reference position, the projection of its location onto a^*, and a map whose axis encodes location should draw each extremum at its reference (Fig. 1). The obstacle is the tree itself. A profile shows the tree exactly only if every subtree occupies one contiguous block of leaves, but the references of two sibling subtrees may interleave (Fig. 2d). No exact profile can then place every feature at its reference, whatever algorithm draws it. This conflict raises three questions: how much displacement the tree forces, where in the tree this displacement arises, and how much merge-level fidelity must be given up to remove it. Without answers, the positional error of a map cannot be attributed to the data or to its layout, and an analyst cannot judge whether a spatial axis is worth its cost.

We answer these questions with bounds and turn the answers into a construction. Three lower bounds, for exact topology, for features of finite width and for a bounded merge error, depend only on the trees, the references and the widths, so the positional price of the axis is known before any layout is computed and applies to every map that keeps the tree. AnchorMTM then computes the map directly from the field sequence and one merge budget ε. It searches leaf orders that the budget admits, using a closed-form test for the smallest merge error an order can show, places the extrema as close to their references as the chosen order allows, and reconstructs the profiles. With $\varepsilon = 0$, the map shows every tree exactly; with $\varepsilon > 0$, it releases the conflicts that a small change of merge levels can resolve and records the merge error each frame actually spends.

Our contributions are as follows.

- We formulate spatially anchored merge tree maps and derive three layout-independent lower bounds on the displacement that keeping the tree forces, for exact topology, finite widths and bounded merge error. The bounds locate the conflict in the tree and measure any merge tree map, including existing ones.
- We present AnchorMTM, which builds the leaf order and extremum positions of every frame from the projected locations, so that the height of a band reads as location and its drift as motion along a^*, while the profile keeps the nesting and merge levels of the tree. A merge budget ε, enforced through optimal merge barriers and checked on every rendered frame, controls how much merge-level fidelity may be traded for position.
- On five planar and two volumetric sequences, we show that the bounds are tight on controlled and real trees, that most of the order error of existing maps is not forced by the tree, and how a budget reshapes the displacement of individual frames.

2 Related Work

2.1 Linearizing Space for Temporal Overviews

Animation preserves the spatial context of each time step, whereas static overviews let viewers compare distant times directly; which serves better depends on the task [45, 51]. Bach et al. describe both,

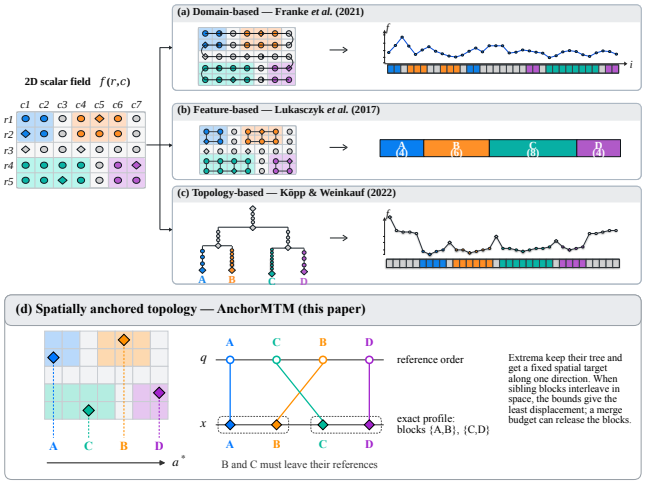


Figure 2: Mapping strategies on a common 35-sample field. (a) Domain traversal keeps samples; (b) a 22-sample feature summary keeps selected structures; (c) a topology-based profile keeps the hierarchy, but its axis has no spatial meaning. (d) AnchorMTM keeps the hierarchy and gives every extremum a target along one direction a^*; here the targets interleave the subtree blocks $\{A,B\}$ and $\{C,D\}$, the conflict that Sections 4 and 5 quantify and resolve.

and the operations between them, as transformations of a space–time cube [2]. A static overview must linearize space, and Fig. 2 contrasts three ways to do so. *Domain-based* linearizations traverse the samples: Hovmöller diagrams keep one physical coordinate and average over the others [25], while geographic orderings, dense pixel displays and space-filling curves visit the whole domain [9, 20, 44], and volume lines and data-driven curves extend such traversals to volumes [57, 64]. *Feature-based* summaries order selected entities: one-dimensional projections, MotionRugs, stable visual summaries and MoReVis rank moving objects by position or stability [6, 18, 52, 60]. Every linearization distorts some spatial relations [1, 39]. These overviews keep a spatial coordinate but no feature hierarchy; AnchorMTM keeps the merge tree and makes explicit what a spatial coordinate then costs.

2.2 Topology of Time-Varying Scalar Fields

Merge trees and contour trees summarize how level sets connect, and persistence-based simplification controls their size [13, 21, 50]. Features are tracked over time by region overlap or attribute similarity [46, 47] and shown as tracking graphs and nested tracking graphs [35, 36, 58]. Time-varying merge trees have been computed and compared with Wasserstein and geometry-aware interleaving distances [40, 42, 61, 63], and such pipelines follow tropical cyclones and atmospheric rivers [31, 62]. The merge error we budget is a labelled cophenetic discrepancy that equals the labelled interleaving distance [19, 37, 38] and relates to distances between ordered merge trees [5]. Tracking graphs show how features appear, split and merge, but not where they are; merge tree maps additionally show the values and nesting of every time step, and AnchorMTM adds their location.

2.3 Drawing Hierarchies along One Axis

Topological landscapes draw a merge tree as a terrain or as a one-dimensional profile whose minima are the extrema and whose barriers are the merge levels [41, 55]; geometry-preserving and temporally coherent variants exist [4, 23], and contour trees have been drawn in the plane and in joint layouts [22, 34]. Temporal treemaps lay out evolving hierarchies along time [29]. TMTM aligns the profiles of successive merge trees into one static map [30], and ST-MTM adds relative spatial distances and temporal stability [10];

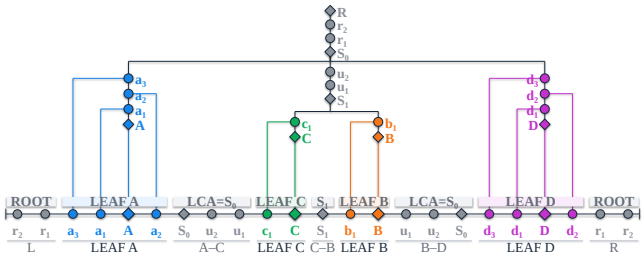


Figure 3: An augmented merge tree and its profile. Extrema and the regular samples of their leaf arcs fill each leaf interval; the gaps between leaves are filled from the paths to their lowest common ancestor (grey), so the largest value between two extrema shows their merge level. These gap fillings are what a relaxed order changes: AnchorMTM sets each gap to an optimal barrier (Proposition 1) instead of the original LCA level.

their axes encode tree traversal or relative distance. Storyline layouts face a related problem for groups of entities and balance crossings and wiggles against group contiguity [33, 49].

The conflict we study arises whenever a hierarchy must respect an external order. Optimal leaf ordering selects a tree-consistent order of a dendrogram that keeps similar leaves adjacent [3, 14], and tree reordering seeks the tree-consistent permutation closest to a target order [7]. Spatially ordered treemaps and data–spatial grid maps trade hierarchy or data layout against geographic position [53, 59], geophylogenies connect leaf orders to map locations [27], and tanglegram layouts minimize crossings or displacement between two trees [17, 26]. Separation constraints keep laid-out objects disjoint [12, 32], minimax ultrametric fitting is closely related to our optimal barriers [16], and merge distortion and topology-preserving projections measure how much hierarchy a mapping retains [11,15]. These methods optimize crossings, sums or stress for a given instance. We instead bound, in closed form for points, the worst displacement that every tree-consistent layout must accept, together with the merge error that removes it.

3 Problem Formulation

3.1 Merge Trees, Profiles, and Feature Identity

For a sampled field f_t, the merge tree T_t records sublevel components created at minima and joined at saddles; negating the field handles maxima [50]. Extraction and persistence simplification precede layout. A leaf i has extremum location p_i^t and support S_i^t, comprising the samples on its leaf arc. Augmented internal arcs retain samples for filling the profile. Write $h_t(v)$ for a node's scalar level and $\mathscr{D}_t(v)$ for its descendant leaves. The fixed-label merge matrix is

$$V_{ij}^t = h_t(\mathrm{lca}(i,j)), \quad i \neq j; \qquad V_{ii}^t = f_t(p_i^t). \tag{1}$$

Equal-level merges form one multifurcation. A one-dimensional profile ψ_t places the extrema in leaf order π_t; the largest profile value between anchors i, j gives their displayed merge level $V_{ij}^{\psi_t}$. Exact topology requires each strict subtree to form one contiguous block.

On a fixed grid, one-to-one Hungarian matching maximizes $|S_i^{t-1} \cap S_j^t|$ and discards zero-overlap pairs [10]; a matched leaf inherits its track identity, and $\mu_t(i)$ denotes its predecessor. These links supply the temporal targets and the motion samples below, and the same matching is reused for every budget (Supplementary Fig. S2).

3.2 Fixed Spatial Reference

The default direction maximizes squared matched-extremum motion:

$$a^* \in \arg\max_{\|a\|=1} a^\top \Big(\sum_{t,i} \Delta p_i^t (\Delta p_i^t)^\top \Big) a, \tag{2}$$

where $\Delta p_i^t = p_i^t - p_{\mu_t(i)}^{t-1}$ over valid links. Its sign makes total projected motion nonnegative; the longest domain axis handles zero motion energy. An analyst may supply a physical direction instead. For nearly zero signed motion, the largest-magnitude direction component is made positive; repeated leading eigenvalues can be resolved using a chosen physical axis. This separates reference estimation from the layout search: a budget changes the admissible map, while the coordinate being approximated stays fixed. Projected domain corners define one origin o and extent L, giving

$$q_i^t = \langle a^*, p_i^t \rangle - o \in [0, L]. \tag{3}$$

Direction, origin and extent remain fixed across frames and budgets. Relative geometry uses support-centroid distances d_{ij}^t; the targets q_i^t use extremum locations. Thus support growth can change relative-distance terms without moving a feature's spatial target.

3.3 A Unified Layout Formulation

Each leaf occupies an interval of width $W_i = s m_i$ and center z_i, where m_i is its support measure and s is fixed. The layout set $\mathscr{L}(\pi; W)$ places these intervals in order π inside $[0, L]$, with gap at least g and anchor slack $|x_i - z_i| \leq \rho W_i/2$. Two order costs distinguish point ordering from packing:

$$\begin{aligned} \mathrm{NROE}(\pi) &= \frac{\max_{j \leq k} [q_{\pi(j)} - q_{\pi(k)}]^+}{2L}, \\ \tau(\pi) &= \min_{(x,z) \in \mathscr{L}(\pi;W)} \|x - q\|_\infty. \end{aligned} \tag{4}$$

NROE is the least normalized worst displacement of points in order π; τ includes the intervals and is an exactly solvable minimax linear program. Merge error measures the displayed hierarchy:

$$d_{\mathrm{top}}(\psi, T) = \max_{i,j} |V_{ij}^{\psi} - V_{ij}|. \tag{5}$$

The diagonal includes extremum values. Under complete fixed labelling this cophenetic error equals labelled interleaving distance and bounds the unlabelled interleaving and bottleneck distances [19, 37, 38].

These variables separate layout priorities in the common model

$$\begin{aligned} \min_{\pi \in \Pi, (x,z) \in \mathscr{L}(\pi;W)} \; & \lambda_R E_R + \lambda_G E_G + \lambda_M E_M + \lambda_C E_C \\ \text{s.t. } & d_{\mathrm{top}} \leq \eta. \end{aligned} \tag{6}$$

Each term is a mean squared residual over the n_t leaves, the ordered pairs P_t or the matched leaves M_t of a frame:

$$\begin{aligned} E_R &= \tfrac{1}{n_t} \textstyle\sum_i (x_i - q_i)^2, & E_G &= \tfrac{1}{|P_t|} \textstyle\sum_{(i,j) \in P_t} (x_j - x_i - d_{ij})^2, \\ E_M &= \tfrac{1}{|M_t|} \textstyle\sum_{i \in M_t} (x_i - \hat{x}_i)^2, & E_C &= \tfrac{1}{n_t} \textstyle\sum_i (x_i - z_i)^2, \end{aligned} \tag{7}$$

with the temporal target $\hat{x}_i$ of Section 5. The allowance η makes the trade-off explicit: $\eta = 0$ demands the exact tree, and a larger η admits orders that follow the references more closely. TMTM packs tree-compatible blocks and aligns them by overlap; ST-MTM fits relative geometry with temporal constraints. Both keep the tree ($\eta = 0$) and have no reference term ($\lambda_R = 0$), so nothing ties their axes to location. AnchorMTM adds E_R and a declared allowance.

3.4 Conflict Between Topology and Spatial Position

A projection can follow every q_i, but a profile that shows the tree exactly must keep each subtree in one block. In Fig. 2d, the tree joins A with B and C with D, while the references occur in the order A,C,B,D: the two blocks interleave, so some extrema must leave their references. Sibling blocks can also nest, as when C and D both lie between A and B (Fig. 5). Keeping the blocks displaces anchors; allowing some merge-level error may admit an order closer to the references. How far the anchors must move depends only on the tree, the references, the widths and the allowance, not on the algorithm that draws the map.

We address three questions: how much displacement exact topology forces (RQ1), how finite feature widths and interval constraints increase that cost (RQ2), and which conflicts can be released by bounded merge-level error (RQ3). The next section answers them in this order; the construction then uses the same quantities to select and place a map.

4 The Positional Price of Keeping the Tree

The positional cost comes from two sources: constraints imposed by the hierarchy and additional choices made by a layout. We first isolate the hierarchy in a point model, then account for interval packing, and finally allow controlled changes to merge levels. This progression answers RQ1–RQ3 using the same reference coordinate. Suppress t, write $[v]^+ = \max(v,0)$ and take empty maxima as zero; complete packing derivations are in the supplementary material.

4.1 Exact Topology

The first question removes feature width and asks what the hierarchy alone forces. Exact topology keeps sibling subtrees in separate blocks, so their reference hulls determine the cost of choosing either block orientation. Let $[m_X, M_X]$ be the reference hull of subtree X in the reduced equal-level tree.

Theorem 1 (Exact-topology position bound)**.** *Every exact profile satisfies* $\mathrm{NROE}(\pi) \geq \nu^*$*, where*

$$\nu^* = \frac{1}{2L} \max_{v} \max_{X \neq Y \in \mathrm{ch}(v)} [\min(M_X - m_Y, M_Y - m_X)]^+. \tag{8}$$

Recursive ordering by child-hull midpoint attains ν^* *in* $O(n \log n)$ *time.*

Sibling blocks must take one orientation. Its inversion cost is either $[M_X - m_Y]^+/2$ or $[M_Y - m_X]^+/2$, and midpoint order chooses the smaller. Every inverted leaf pair separates at its LCA. Therefore error above ν^* can be removed by reordering in the point model.

4.2 Finite Widths

Set $H_i = \rho W_i/2$, $\ell_i = q_i - W_i/2 - H_i$, $h_i = q_i + W_i/2 + H_i$ and $s_i = W_i + g$. Packing summaries

$$\begin{aligned} A(S) &= \max_{a \in S} \big(\ell_a + \sum_{i \in S : \ell_i \geq \ell_a} s_i\big), \\ B(S) &= \max_{b \in S} \big(\sum_{i \in S : h_i \leq h_b} s_i - h_b\big) \end{aligned} \tag{9}$$

measure the interval span forced by upper and lower reference-endpoint sets. Define $\omega(X,Y) = \frac{1}{2}\min\{A(X) + B(Y), A(Y) + B(X)\} - g/2$, and $E_\partial = \max\{0, B(\mathscr{F}) - g, A(\mathscr{F}) - g - L\}$ for all leaves $\mathscr{F}$.

Theorem 2 (Width-aware position bound)**.** *For every feasible exact-topology interval layout,*

$$\|x - q\|_\infty \geq \tau(\pi) \geq \mathscr{B}_w = \max\{E_\partial, \max_{(X,Y) \in \mathscr{P}_0} \omega(X,Y)\}, \tag{10}$$

where $\mathscr{P}_0$ *contains distinct sibling descendant-leaf sets. With zero widths and gaps,* $\mathscr{B}_w = L\nu^*$*.*

The first and last anchors must accommodate their intervals and intervening gaps up to the permitted anchor slack. Sibling contiguity chooses a block orientation, while the canvas ends supply E_∂. Thus ordering alone no longer determines the positional cost: even an order attaining the point bound may require additional displacement to fit its intervals. The bound has coordinate units, while ν^* is normalized by L. This motivates evaluating the exact packing cost of each candidate order before optimizing its anchors.

4.3 Bounded Merge Error

If leaf k lies between descendants i, j of node u, interval containment gives $V^{\Psi}_{ik} \leq V^{\Psi}_{ij}$, hence $V_{ik} \leq h(u) + 2\eta$. Let $A_\eta(u)$ be the highest ancestor with $h(A_\eta(u)) - h(u) \leq 2\eta$. Form $\mathscr{P}_\eta$ from pairs $(\mathscr{D}(u), \{k\})$ with $k \notin \mathscr{D}(A_\eta(u))$, and pairs $(\mathscr{D}(u), \mathscr{D}(w))$ with each set disjoint from the other's A_η ancestor set, for internal u, w.

Theorem 3 (Bounded-topology position bound)**.** *If* $d_{\mathrm{top}} \leq \eta$*, every pair in* $\mathscr{P}_\eta$ *lies wholly in one block orientation. Thus*

$$\begin{aligned} \mathrm{NROE}(\pi) \geq C_\eta &= \max_{(X,Y) \in \mathscr{P}_\eta} \frac{[\min(M_X - m_Y, M_Y - m_X)]^+}{2L}, \\ \tau(\pi) \geq C_{w,\eta} &= \max\{E_\partial, \max_{(X,Y) \in \mathscr{P}_\eta} \omega(X,Y)\}. \end{aligned} \tag{11}$$

The accumulated ancestor gap determines which outside leaves may enter a block, so an allowance releases only the conflicts supported by sufficiently small merge-level differences. Flattening parent gaps of at most 2η gives a second certificate; we use the larger of the two valid bounds. Together, the three results distinguish forced displacement from displacement chosen by a layout. Their algorithmic implication is to search orders using the exact fixed-order cost τ, admit them only when their merge error fits the allowance, and limit additional displacement during anchor placement. A certificate is a lower bound; the construction must still produce and check a feasible map.

5 AnchorMTM

Figure 4 summarizes the construction; its topology-budgeted layout panel contains the separate order-search and anchor-placement stages. Extract and Reference prepares the trees, identities and fixed targets defined in Section 3. Search Leaf Orders chooses an admissible order with low τ. Place Anchors balances spatial and temporal objectives within a cap on that order's displacement. Reconstruct and Assemble builds the scalar profiles and stacks them along time. The output of each stage fixes the constraints passed to the next. The control is a normalized merge allowance ε, with $\eta = (\varepsilon_0 + \varepsilon)\Delta_F$ for sequence scalar range Δ_F and numerical tolerance $\varepsilon_0 = 10^{-7}$. Exact Mode sets $\varepsilon = 0$; the exact experiments use zero labelled error. Budgeted Mode uses a positive allowance within the same construction. Balanced denotes an experimentally selected positive-budget configuration, not all budgeted maps. Widths, gap, slack and weights stay fixed during a budget sweep; the merge allowance controls hierarchical error rather than directly bounding positional error.

5.1 Order Selection and Optimal Barriers

Order search receives the prepared tree, reference targets, widths and merge allowance. A finite candidate pool $\mathscr{C}$ starts from midpoint orders of the strict tree, threshold flattenings and selective conflict flattenings. Width-aware block moves refine promising structures. The point bound supplies an exact-topology initialization, while the interval analysis motivates evaluating packing displacement. Each visited order is checked against the original merge

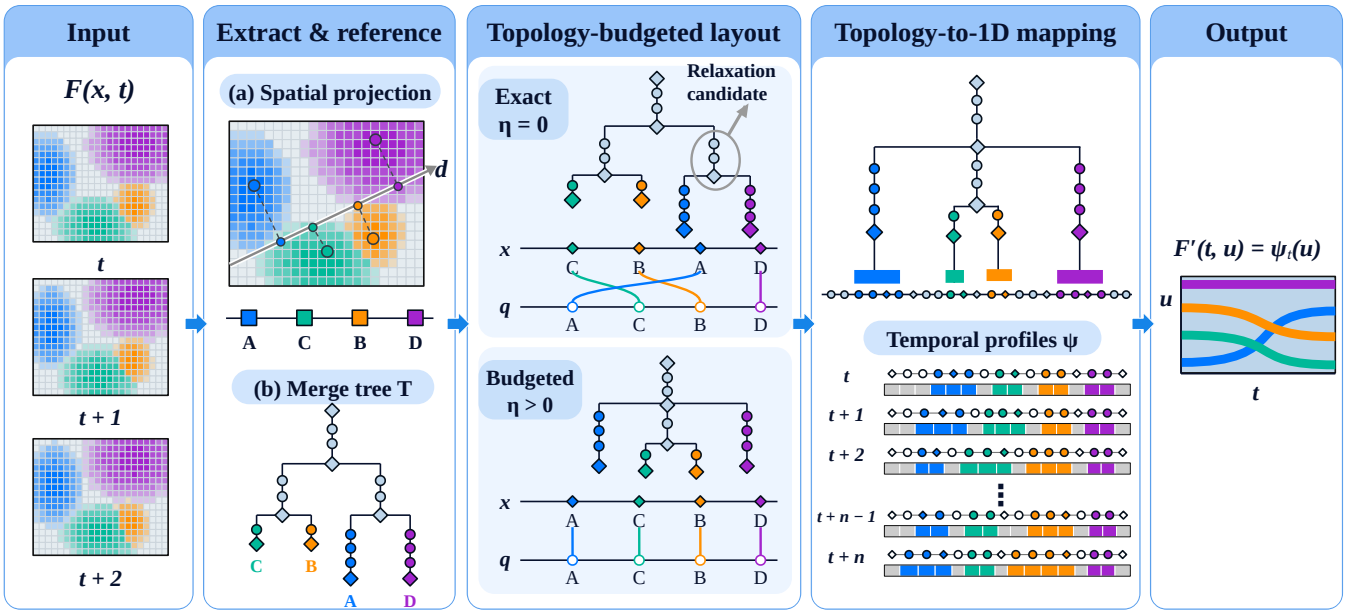


Figure 4: AnchorMTM workflow on a four-leaf example. Extract and reference: merge trees of the input frames $F(x,t)$, leaf supports matched across frames, and one direction $d = a^*$ that gives every extremum a reference position q; the bounds follow from these quantities alone. Topology-budgeted layout: with $\eta = 0$ the leaf order keeps every subtree contiguous and the anchors x leave their references; with $\eta > 0$ a relaxation candidate admits an order that follows them, and the anchors are placed under a displacement cap. Topology-to-1D mapping: leaf-arc samples and optimal barriers fill each profile ψ_t, every rendered frame is checked against the budget, and the stacked profiles form the map.

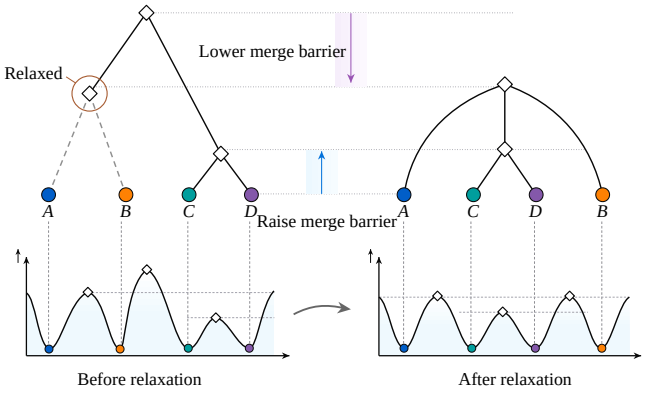


Figure 5: Selective flattening admits order A,C,D,B while retaining the C,D grouping. The displayed barrier levels are corrected against the original merge matrix.

matrix, and selection is lexicographic:

$$\pi^{\varepsilon} \in \arg\min_{\pi\in\mathscr{C}} \big(\tau(\pi), \mathrm{NROE}(\pi)\big) \quad \text{s.t.} \quad \delta^*(\pi) \le \eta. \tag{12}$$

The exact tree supplies a feasible strict-topology seed. Selective flattening releases additional permutations while retaining deeper blocks, as illustrated in Fig. 5.

Construction sequence. For each frame, generate and refine candidate orders; compute τ, NROE and δ^*; admit the budget-feasible orders; choose the lexicographic minimum; solve its capped QP; fill and correct the profile; then check the rendered merge matrix and append the temporal column. The supplementary algorithm lists these steps with the tracking and rounding procedures.

For gap k of an order, set $C_k = \min_{i\le k<j} V_{\pi(i)\pi(j)}$ and let F_k be the larger adjacent extremum value.

Proposition 1 (Optimal barriers)**.** *With fixed extrema, the least continuous merge error is*

$$\delta^*(\pi) = \max\Big\{0, \max_k (F_k - C_k), \tfrac{1}{2}\max_{i<j}\big(V_{\pi(i)\pi(j)} - \max_{i\le k<j} C_k\big)\Big\}, \tag{13}$$

attained by $b_k = C_k + \delta^*(\pi)$.

Indeed, error δ requires $F_k \le b_k \le C_k + \delta$; choosing the largest permitted barriers meets every pair's lower requirement exactly when $\delta \ge \delta^*$. This gives a closed-form admission test for candidate orders.

5.2 Capped Spatiotemporal Placement

For the chosen order, the default weights are $(\lambda_R, \lambda_G, \lambda_M, \lambda_C) = (4, 1, 0.5, 0.1)$. Each term in Eq. 6 is a mean squared residual: anchor-to-reference distance, ordered anchor separation minus support-centroid distance, matched temporal-target error, or anchor-to-center offset. The temporal target follows projected motion:

$$\hat{x}_i^t = x_{\mu_t(i)}^{t-1} + q_i^t - q_{\mu_t(i)}^{t-1}. \tag{14}$$

The selected order is now fixed, so placement changes coordinates without revisiting the discrete search. Unmatched leaves have no temporal penalty. Placement solves the convex quadratic objective under the interval constraints and

$$\|x - q\|_\infty \le \tau(\pi^{\varepsilon}) + \beta_0, \tag{15}$$

with fixed slack β_0. Because τ is attained, this cap is feasible and transfers the selected order's displacement control to the final anchors. The cap prevents a relative-distance or temporal term from moving an anchor arbitrarily far from its spatial target. Conversely, the fixed slack leaves room to improve these secondary relationships after the discrete order has been chosen. Within a frame the feasible set is linear and the objective quadratic; frame order enters only through matched predecessor targets.

Table 1: Five-dataset comparison. Bold/underline mark best/second distinct values among comparable local measurements only; ties share emphasis. Daggered SNS/TW values are published context and excluded from ranking. Curves use fixed default orientations.

Dataset	Hilbert			Morton			TMTM			ST-MTM			AnchorMTM (ours)			
	SNS↓	TW↑	NPE↓	SNS↓	TW↑	NPE↓	SNS↓	TW↑	NPE↓	SNS↓	TW↑	NPE↓	Mode	SNS↓	TW↑	NPE↓
Gaussian	0.1117	0.6719	0.7557	0.0393	0.8333	0.7359	0.25†	0.77†	0.1118	0.06†	1.00†	0.1154	Exact	**0.0353**	**0.9323**	**0.0089**
													Balanced	**0.0353**	**0.9323**	**0.0089**
Ring	**0.1252**	**0.9048**	0.5941	0.1489	0.7619	0.6033	0.28†	0.68†	0.1756	0.21†	0.88†	0.2463	Exact	0.1485	0.8571	0.0471
													Balanced	0.1334	0.8571	**0.0395**
Wildfires	0.1883	0.8303	0.3641	0.1304	0.8758	0.1560	0.44†	0.88†	0.2280	0.24†	0.91†	0.3180	Exact	**0.0426**	**0.9136**	0.0131
													Balanced	0.0447	0.8989	**0.0116**
Storms	0.1290	0.7403	0.3705	0.1180	0.7251	0.3468	0.34†	0.71†	0.5753	0.21†	0.94†	0.4579	Exact	0.0790	0.8463	0.0399
													Balanced	**0.0743**	**0.8766**	**0.0140**
PM2.5	0.2386	**0.8887**	0.8090	0.1760	0.8277	0.8823	0.3018	0.7781	0.5241	0.1897	0.8224	0.5577	Exact	0.1964	0.7884	0.1390
													Balanced	**0.1636**	0.7657	**0.0468**

For the original four inputs, TMTM/ST-MTM Temporal SNS/TW (†) are published values; other entries are local measurements. Author inputs and coverage are unconfirmed. Local TW coverage (Gaussian/Ring/Wildfires/Storms) is 64/64, 1/40, 4/32, 11/744; ineligible frames are omitted. Published TW has two-decimal precision. Mode applies only to AnchorMTM. PM2.5 uses 744 shared-input local frames (TW $k = 1$, full coverage); its Balanced mode uses the fixed exploratory 5% case budget, without validation-set budget selection. ST-MTM uses the Storms preset with start frame 0; PM2.5 baseline metrics are all recomputed.

5.3 Profile Reconstruction

The final stage receives the order, anchor intervals and optimal barriers. Leaf-arc samples fill the intervals and LCA paths fill their gaps [10, 30]. Barrier correction clips each inter-anchor interior to b_k and raises its peak when needed. Profiles share direction, origin, extent and pixel scale; their stack is $I^{\varepsilon}(t,u) = \psi_t^{\varepsilon}(u)$. Integer rounding can affect merge levels, so the rendered fixed-label matrix, including its diagonal, is checked after reconstruction. The supplementary material supplies the full algorithm and rounding procedure.

6 Evaluation

The evaluation follows the argument of the paper. We first test the bounds and use them to measure how much of the positional error of existing maps the tree actually forces; this evidence does not depend on what AnchorMTM optimizes. We then compare where the maps draw features, examine how a merge budget changes individual frames, and read two maps against their source fields.

6.1 Data and Measures

Data. The five planar sequences are Gaussian (64 frames), Ring (40), Wildfires (32) and Storms (744), the inputs of the ST-MTM benchmark [10] reconstructed with its per-frame persistence fractions, and hourly $PM_{2.5}$ over northern China in January 2017 (744 frames, 5% persistence) [28]; Storms uses ERA5 sea-level-pressure anomalies [24] and Wildfires aggregated fire radiative power [18]. The volumes are the flow past a square cylinder (102 frames of the Q-criterion, 2% persistence) [8, 56] and the Tangaroa ship wake (201 speed fields, absolute persistence 0.2) [43]. Within each comparison, all methods share the fields, trees, identities and references.

Measures. Position is measured by NPE, the worst displacement of a frame relative to the axis length, averaged over frames: $\mathrm{NPE} = \frac{1}{T}\sum_t \max_i |x_i^t - q_i^t|/L$. Relative geometry is measured by scale-normalized stress (SNS) [48] and trustworthiness (TW) [54] of anchor distances against support-centroid distances, the goal of ST-MTM; for the four benchmark inputs, the published TMTM and ST-MTM values are quoted for context (daggers in Table 1). Neither baseline fixes the direction of its axis, so each is reflected once per sequence to agree with the reference; Hilbert and Morton curves use fixed orientations.

Modes. Exact uses zero allowance. Balanced uses a positive budget selected once from 1–10% on the first chronological quarter of each benchmark input (1, 1, 5 and 10%); $PM_{2.5}$ uses a fixed 5%. The volumetric table uses fixed $+x$ references and stored configurations, and the volume figure one default configuration; the supplement lists every setting.

6.2 Tightness of the Bounds and Forced Error

Controlled trees. We first check the bounds where the optimum is known. Two sibling pairs merge below a root, with references $(0.2, 0.5, 0.5-d, 0.8-d)$; across 43 overlaps d, independent minimax linear programs over all 24 leaf orders, the width-aware bound and the selected order coincide at $\tau = \max(0, (d+0.04)/2)$, and for four parent gaps h all 364 budget tests release the conflict exactly at $2\varepsilon = h$, as Theorem 3 predicts (Supplementary Fig. S5).

Real trees. The selected Exact order attains the width-aware bound on 1,335 of 1,624 planar frames, so on these frames no exact-topology order has a smaller worst displacement. Gaussian and Ring attain it on every frame, Wildfires misses one frame and Storms 15. $PM_{2.5}$ attains it on 471 of 744 frames and leaves a small residual gap: the mean interval cost is 13.11% of the axis against a mean bound of 12.88%. Volumes are harder for the heuristic search; the supplement lists their attainment.

Forced and chosen error. Because midpoint ordering attains ν^* on the same tree, the bound splits the order error of any exact-topology map into a part the tree forces and a part its order rule chooses (Fig. 6). On the real-data sequences, 83–98% of the order error of TMTM and ST-MTM on Wildfires and Storms and 56–60% on $PM_{2.5}$ lie above ν^* and could be removed without changing the tree; on the two volumes the share is 31–54%. The bounds also reveal how strongly a field entangles its hierarchy with space: ν^* is 0.16% of the axis on Wildfires and 2.6% on Storms, but 12.5% on $PM_{2.5}$, where nested pollution maxima interleave along the chosen direction. The edge traced by AnchorMTM shows what a budget releases. Where conflicting subtrees merge just below their parent, little is needed: 1% removes almost all forced order error on Wildfires (0.16% to 0.01%) and on Cylinder (12.2% to 0.2%). Where conflicts lie deep, position is expensive: $PM_{2.5}$ needs 10% to fall below 1%, and Tangaroa keeps more than 7% up to $\varepsilon = 5\%$. The projection reaches zero order error only by changing merge levels by 4–62% of the scalar range.

6.3 Absolute Spatial Positioning

At the same zero merge error, Exact lowers NPE relative to TMTM and ST-MTM by 73–96% on all five planar inputs and is also below the Hilbert and Morton curves, which break the tree (Table 1). Balanced, with a budget of at most 10%, lowers NPE further on Ring, Wildfires and Storms and leaves Gaussian unchanged. NPE is the quantity AnchorMTM is designed for, so these results show what an anchored axis delivers rather than a general ranking. The

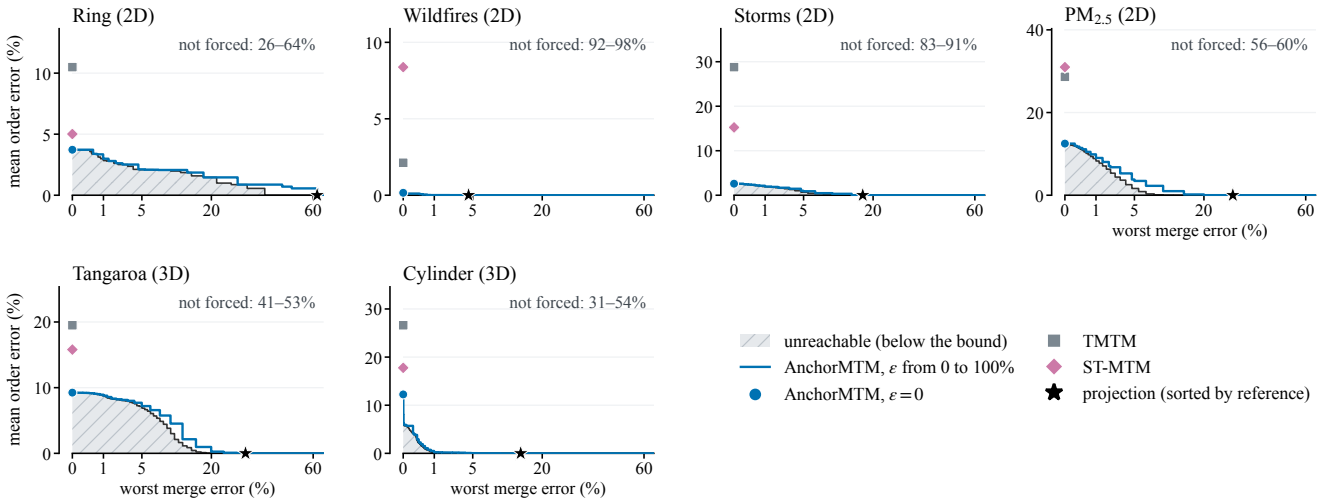


Figure 6: Position–topology plane. Each map is one point, placed by its worst merge error over all frames (square-root scale) and its mean order error. No map can enter the hatched region below the bound (Theorem 3 and the flattened certificate; Theorem 1 at zero merge error). TMTM and ST-MTM keep the tree and lie at zero merge error, far above the bound; the label gives the share of their order error that the tree does not force. The projection follows every reference and breaks the tree. AnchorMTM (blue line, ε from 0 to 100%) runs along the edge of the unreachable region. Gaussian is omitted: its bound is zero.

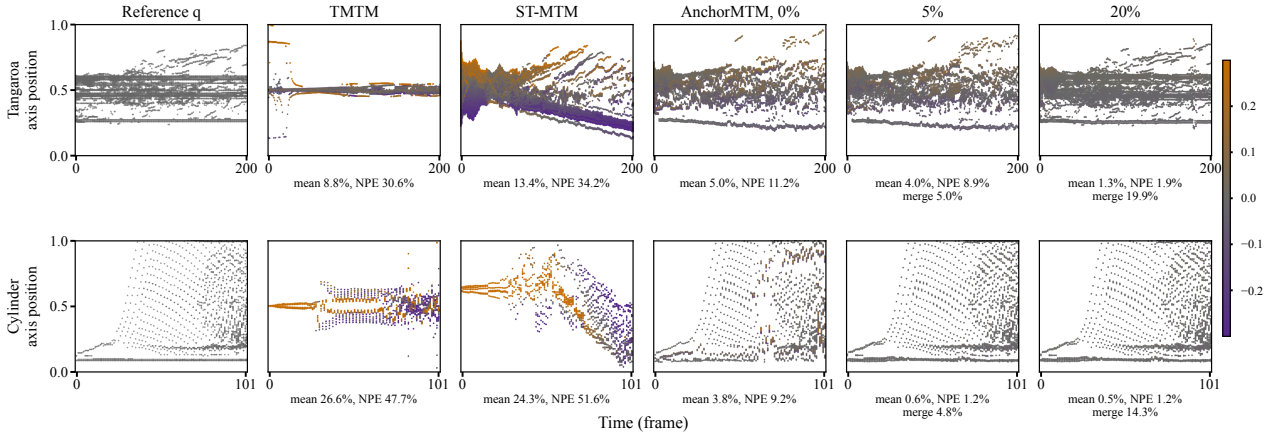


Figure 7: Recomputed feature positions on Tangaroa and Cylinder using the default-weight volume configuration. Grey reference points and signed-displacement colors connect feature positions to q. Each panel reports the feature-pooled mean absolute displacement and NPE (mean per-frame maximum); positive budgets also report maximum attained merge error. Baseline reflection minimizes pooled reference displacement. These runs use the fixed configurations described in the protocol, distinct from the selected geometry-oriented table settings.

other measures show its cost on some inputs. Hilbert keeps better SNS and TW on Ring (0.1252 and 0.9048 against 0.1334 and 0.8571) and better TW on $PM_{2.5}$ (0.8887 against 0.7657), and on $PM_{2.5}$, where every score is measured locally, ST-MTM keeps a lower SNS and a higher TW than Exact (0.1897 and 0.8224 against 0.1964 and 0.7884).

On the volumes, the selected fixed-x configurations lower SNS and NPE and raise TW relative to TMTM (Supplementary Table S4): Cylinder NPE falls from 47.69% to 4.58% and Tangaroa from 38.66% to 6.53%. The Tangaroa configuration has zero reference weight and a 33.33% allowance, so there the gain comes mainly from geometry fitting within the displacement cap. Figure 7 shows the positions behind these numbers under one default configuration. TMTM gathers the Cylinder vortices near the middle of the axis, whereas Exact already reproduces the shedding pattern of the references (NPE 9.18% against 47.69%), and a 5% budget removes almost all of the rest (1.19%). The conflicts of Tangaroa lie deeper: Exact lowers NPE from 30.63% to 11.18%, and the pattern of the references appears only near $\varepsilon = 20\%$ (1.88%).

6.4 Where a Merge Budget Acts

A budget limits merge error, not displacement, so its effect depends on how many frames contain a conflict it can release (Fig. 8). Two regimes appear. On Storms the budget acts on a tail: 35% of the frames use the allowance at $\varepsilon = 10\%$, the median displacement moves only from 1.10% to 0.84%, but the 90th percentile falls from 10.7% to 2.5% and the worst frame from 44.8% to 11.6%. On $PM_{2.5}$ and the volumes almost every frame contains a releasable conflict and the whole distribution moves: on $PM_{2.5}$, 97% of the frames use a 5% budget and the median falls from 14.5% to 2.1%; on Cylinder, 2% already lowers the median from 6.8% to 1.0%; on Tangaroa the median falls from 11.3% to 5.9% at 10% and to 1.2% at 20%. Ring lies in between: its median falls from 3.9% to 1.0% at 5%, while its 90th percentile stays at 8.5% up to 20%. The share of frames that use the budget is thus itself a diagnostic of how strongly a sequence entangles hierarchy and space. Exact

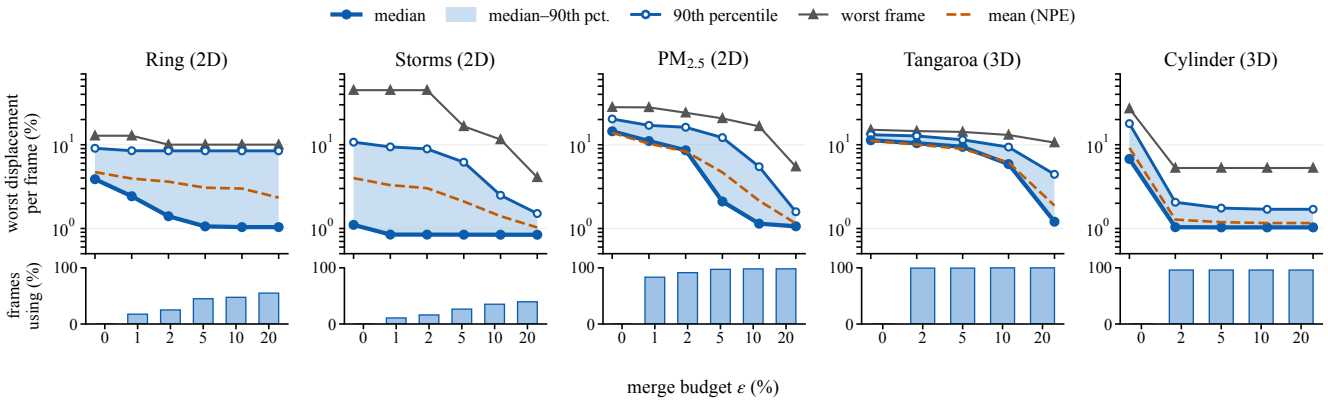


Figure 8: Where a merge budget acts. For each sequence and budget: the median, 90th percentile and worst frame of the per-frame worst displacement (log scale), and the mean (NPE, dashed). Below: share of frames whose attained merge error is positive. Every frame stays within its budget.

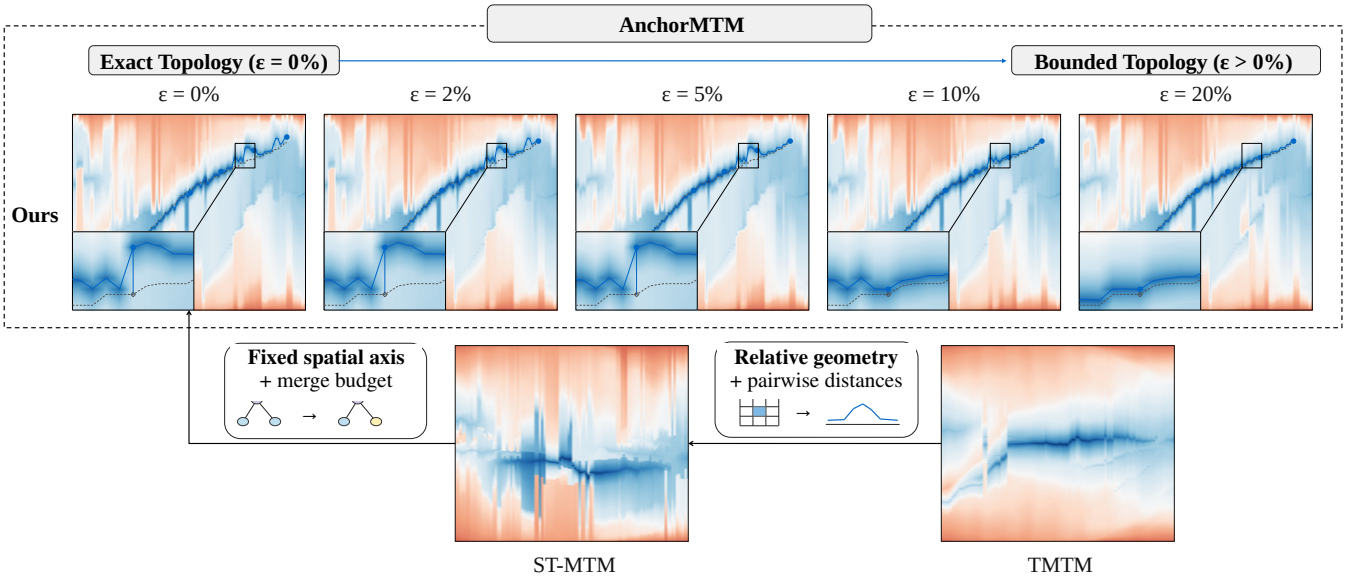


Figure 9: Storms event window, frames 355–460, drawn by AnchorMTM at five budgets and by TMTM and ST-MTM on the same fields and trees, on one $[-35, 35]$ hPa scale. The boxes state what each map adds to the one on its right: ST-MTM adds pairwise spatial distances to TMTM's tree order, and AnchorMTM gives the axis one fixed direction and controls the merge levels it may change. The dashed curve in the insets is the reference trajectory of the tracked storm.

has zero merge error on every input; Balanced attains at most 0, 0.982, 4.421 and 9.985% on Gaussian, Ring, Wildfires and Storms, and every frame passes the continuous and raster checks. Relative geometry can move either way: Wildfires SNS rises from 0.0426 to 0.0447, and $PM_{2.5}$ TW falls from 0.7884 at $\varepsilon = 0$ to 0.7580 at 20%.

6.5 Reading Anchored Maps

A storm crossing the North Atlantic. Figures 1 and 9 follow one low from the open Atlantic to Scandinavia between 16 and 19 December 1999. In the AnchorMTM maps its band rises steadily along the axis, and the rise is the storm's eastward motion, while the profile still shows how it merges with the neighboring lows. TMTM aligns subtrees by temporal overlap and ST-MTM places them by relative distance, so both keep the storm near the same height and its motion cannot be read from the map. The budget changes the picture locally: at 0, 2 and 5% the maps coincide in this window, while 10 and 20% release the remaining conflict and move the storm onto its reference trajectory (insets).

Vortex shedding behind a cylinder. Figure 10 reads the wake of a square cylinder in a variant that guides the layout toward a central tree. Five stages separate clearly: an early, nearly steady wake (I), a transition (II), regular shedding (III), a second transition (IV) and the late wake (V), each matched by the source isosurfaces. The variant trades location for hierarchical unfolding: its ordinate is no longer the streamwise coordinate, and the frame-balanced streamwise error rises from 21.1% to 28.1% as the guidance weight grows from 1 to 16. A tree-ordered TMTM map shows the same stages (Supplementary Fig. S8); the anchored default configuration instead places the shed vortices at their streamwise positions (Fig. 7). The two readings are complementary: one shows when the wake changes regime, the other where each vortex is.

6.6 Implementation and Verification

Every figure is regenerated from labelled layouts and raster columns: each rendered column is checked against the fixed-label merge matrix of its source tree, together with interval order, capacity, gaps and anchor slack, and the bounds and baseline inversion er-

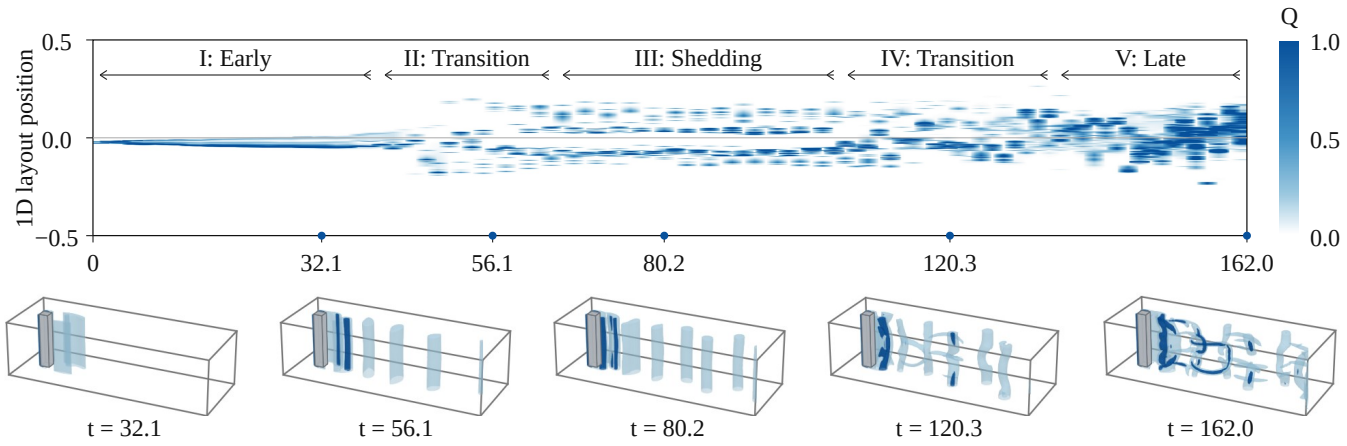


Figure 10: Flow past a square cylinder (102 frames of the Q-criterion) in a central-tree guidance variant of AnchorMTM, with five source isosurfaces at a common camera and blue scalar scale. Stages I–V mark the early wake, the onset of shedding, regular shedding, a second transition and the late wake. In this variant the ordinate is a guided layout position rather than the streamwise coordinate.

rors are recomputed independently (Supplementary Table S3). Ordering and placement take 0.15–3.32 s per benchmark sequence and 200 s for the 744-frame $PM_{2.5}$ sequence on an Apple M4; the supplement adds full pipeline times, the Ring guidance study and the $PM_{2.5}$ episode in three maps.

7 Discussion

What the bound reveals about the data. The exact-topology bound is a property of a field and a direction, not of a layout, and it is informative on its own. A small ν^*, as on Wildfires and Storms, says that the hierarchy of the field and its geography largely agree: sibling subtrees occupy separate stretches of the axis, and any good order can draw features near their locations. A large ν^*, as on hourly $PM_{2.5}$, says that nested maxima interleave in space; there, a map that keeps the tree must displace features, and the displacement measures how far the merge structure of the field departs from its spatial arrangement. Because the bound requires no layout, it can be computed for several candidate directions before a map is drawn and used to choose the axis along which hierarchy and space conflict least.

Forced error versus layout choice. The gap between existing maps and the bound is the most practical finding. Most of the order error of TMTM and ST-MTM on our real-data sequences is not forced by the tree; it follows from ordering leaves without a spatial reference. Any merge tree map that keeps the tree, including future ones, can be measured against the same bound and judged on how much of its displacement it chooses.

Spending topological fidelity deliberately. A merge budget changes merge levels by a declared amount; it never removes a feature, alters an extremum value or moves the data. Persistence simplification at a comparable threshold would delete features below it instead. The per-frame records show where the budget was spent, and the share of frames that use it distinguishes sequences whose conflicts are rare and deep, such as Storms, from sequences in which nearly every frame is entangled, such as $PM_{2.5}$ and the volumes. Marking the changed barriers in the map can keep this trade visible to the reader.

One axis, several readings. A fixed spatial axis and a geometry-oriented embedding answer different questions. AnchorMTM supports reading position and motion along a chosen direction, ST-MTM preserves local neighborhoods better, and the central-tree variant of Fig. 10 exposes regime changes. These readings complement one another, and a view that links them, or small multiples along several directions, could serve analysts who need more than one.

Limitations and future work. AnchorMTM encodes one spatial direction; positions across it, and the full two- or three-dimensional geometry, are not shown on the axis. We plan to extend the bounds from one axis to a complete account of the fidelities a temporal map trades against one another, including position along several directions, temporal coherence and relative geometry, each with its own certificate. The default direction is estimated from the whole sequence, so streaming data need a direction chosen in advance. The order search is heuristic: it attains the width-aware bound on most planar frames but leaves a residual gap on $PM_{2.5}$ and on large volumetric trees. Finally, we have not yet measured how accurately analysts read position, motion and nesting from these maps; a controlled user study is the next step.

8 Conclusion

We presented AnchorMTM, a merge tree map whose vertical axis is one fixed spatial direction, so that an analyst can read where features are and how they move while seeing how they nest. Keeping the merge tree and keeping location conflict, and we addressed this conflict in three steps: we formulated the construction of a spatially anchored map as one layout problem under a declared merge-error limit; we derived three lower bounds on the displacement that the tree forces, for exact topology, finite widths and bounded merge error; and we built the map from the data with a budgeted leaf-order search, optimal merge barriers and a capped placement. On five planar and two volumetric sequences, the selected orders attained the width-aware bound on 1,335 of 1,624 planar frames, most of the order error of existing maps proved not forced by the tree, and at the same exact topology AnchorMTM lowered the mean worst displacement by 73–96%. Because the bounds depend only on the data, they tell before a map is drawn how much a spatial axis costs, where in the tree that cost arises, and how much merge-level fidelity buys it back. This turns the trade-off between topology and position into a quantity that can be measured and chosen, and it lays a foundation for temporal maps that certify several fidelities at once.

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